



docker images Command

Introduction

The **docker images** command is used to **list and manage Docker images** stored on a system.

- What are Docker images?
- How do you use **docker images**?
- How can you manage images efficiently?

This guide covers **how to list, filter, and remove images using docker images**.

Section 1: Learn

What is a Docker Image?

A **Docker image** is a **lightweight, standalone package** that contains:

- Application code
- System libraries and dependencies
- Configuration files

Images are used to **create containers**.

Why Use **docker images?**

Feature	Benefit
Lists available images	Shows images stored locally
Displays image details	Provides size, tags, and creation time



Feature	Benefit
Helps manage disk space	Identifies old or unused images
Supports filtering	Lists images based on name, tag, or creation date

Interesting Fact

Docker images are **layered** and **immutable**, meaning changes create a **new layer** rather than modifying the existing image.

Section 2: Practice

1. Listing All Available Docker Images

```
docker images
```

Example Output:

```
REPOSITORY TAG    IMAGE ID    CREATED     SIZE
nginx      latest  a234bc56def7 2 weeks ago 23MB
ubuntu     20.04   b674ac12d3f9 1 month ago 29MB
```

This shows:

- Repository name (nginx, ubuntu)
 - Tag (version of the image)
 - Image ID (unique identifier)
 - Created date and image size
-



2. Filtering Images

List Images from a Specific Repository

```
docker images nginx
```

This shows **only the Nginx images**.

List Images by Tag

```
docker images --filter "reference=ubuntu:20.04"
```

This shows **only the Ubuntu 20.04 image**.

3. Viewing Image History

```
docker history nginx
```

This displays all **layers and modifications** in an image.

4. Removing an Image

Remove a Specific Image by ID

```
docker rmi <image_id>
```

Remove All Unused Images

```
docker image prune
```



Section 3: Know More

Frequently Asked Questions

1. What is the difference between **docker images** and **docker ps**?
 - o **docker images** lists **available images**.
 - o **docker ps** lists **running containers**.
2. Can I rename a Docker image?

```
docker tag old_image new_repository:new_tag
```

3. How do I download a new image?

```
docker pull nginx
```

4. How do I find large images to free up space?

```
docker images --format "{{.Repository}}: {{.Size}}"
```

5. What happens if I delete an image that a container is using?
 - o The container continues running, but you **cannot create new containers** from the deleted image.

Conclusion

The **docker images** command **helps list, manage, and optimize Docker images**.

Start by **listing images, filtering results, checking history, and cleaning up unused images**, and soon you'll be **efficiently managing Docker environments**!